

Problem-Solving and Programming Completed SPARK Reference

BlueWave Island Tours - question-by-question completed guide

REFERENCE EXAMPLE ONLY

Use this document to study the wording, steps, structure and quality of completed work. Do not submit, rename or copy this SPARK project as your own SBA.

Project: BlueWave Island Tours

Sector: Tourism

Questions being answered

Task A - Problem Definition and Algorithm

1. Write a clear problem statement for the following task: Calculate the final cost and balance for each tour booking and the combined value of bookings processed.

How SPARK completed it

- State what information the program receives.
- State the main calculation or decision the program performs.
- State the result the program must display.

Completed result

Problem statement: Calculate the final cost and balance for each tour booking and the combined value of bookings processed.

2. Identify the inputs, processes and outputs, then write pseudocode containing at least one selection and one repetition statement.

How SPARK completed it

- List inputs, processes and outputs before writing pseudocode.
- Use indentation to show the selection and loop structure.
- Keep variable names consistent from the algorithm through to the program.

Completed result

Inputs: Customer name, Package code, Adults, Children, Amount paid

Processes: Select adult and child rates, Calculate booking cost, Apply 8% group discount when five or more persons are booked, Subtract amount paid, Accumulate booking value

Outputs: Customer summary, Discount, Final cost, Balance, Total booking value

Pseudocode:

START

SET totalValue <- 0

INPUT bookingCount

FOR booking <- 1 TO bookingCount

INPUT customerName, packageCode, adults, children, amountPaid

SET adultRate and childRate for packageCode

SET cost <- (adults * adultRate) + (children * childRate)

IF adults + children >= 5 THEN

SET discount <- cost * 0.08

ELSE

SET discount <- 0

ENDIF

SET finalCost <- cost - discount

SET balance <- finalCost - amountPaid

SET totalValue <- totalValue + finalCost

DISPLAY customerName, discount, finalCost, balance

NEXT booking

DISPLAY totalValue

STOP

Task B - Trace Table and Test Data

1. Create a trace table using at least three sets of test data. Include normal data, a boundary case and an invalid or unusual case where appropriate.

How SPARK completed it

- Choose values that exercise each branch of the selection statement.
- Follow the pseudocode in order and record important variable values.
- Calculate the expected output before running the program.

Completed result

The completed SPARK trace table includes the project's supplied normal examples and adds a boundary/validation discussion in the completed PDF.

Task C - Program Implementation

1. Implement the algorithm in the programming language selected by your centre. The SPARK Pascal reference is saved as TourBookings_YourFullName.pas.

How SPARK completed it

- Create variables using suitable data types.
- Translate each pseudocode step into program statements in the same logical order.
- Compile or run often so errors are found in small sections.
- Use the same test data from the trace table.

Completed result

The SPARK reference implementation is TourBookings_YourFullName.pas. The completed PDF includes the full source code.

2. Test the program and capture evidence that the expected and actual results agree.

How SPARK completed it

- Run normal test data first.
- Run the boundary case.
- Run an invalid or unusual input when the program includes validation.
- Capture screenshots showing both the entered data and the program result.

Completed result

The completed SPARK reference compares expected trace-table outputs with the program output and records the result of each test.

Task D - Documentation

1. Create ProgramDocumentation_YourFullName.pdf containing the problem definition, algorithm, source code, trace table, test data and screenshots of program execution.

How SPARK completed it

- Create a cover page and table of contents.

- Keep the section order the same as the task sequence.
- Use readable code formatting and captions for screenshots.
- Export the final document to PDF and check every page.

Completed result

Completed documentation filename: ProgramDocumentation_YourFullName.pdf. The SPARK section PDF demonstrates the required order and evidence.

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Problem definition

Calculate the final cost and balance for each tour booking and the combined value of bookings processed.

Input - Process - Output

Inputs	Processes	Outputs
Customer name Package code Adults Children Amount paid	Select adult and child rates Calculate booking cost Apply 8% group discount when five or more persons are booked Subtract amount paid Accumulate booking value	Customer summary Discount Final cost Balance Total booking value

Pseudocode

```
START
SET totalValue <- 0
INPUT bookingCount
FOR booking <- 1 TO bookingCount
    INPUT customerName, packageCode, adults, children, amountPaid
    SET adultRate and childRate for packageCode
    SET cost <- (adults * adultRate) + (children * childRate)
    IF adults + children >= 5 THEN
        SET discount <- cost * 0.08
    ELSE
        SET discount <- 0
    ENDIF
    SET finalCost <- cost - discount
    SET balance <- finalCost - amountPaid
    SET totalValue <- totalValue + finalCost
    DISPLAY customerName, discount, finalCost, balance
NEXT booking
DISPLAY totalValue
STOP
```

Trace table and test data

Name	Code	Adults	Children	Paid	Discount	Final Cost	Balance
Nadia Cole	CUL	2	1	5000	0	14500	9500
Eric Stone	ADV	4	2	12000	2400	27600	15600
Maya Singh	NAT	1	0	4000	0	7000	3000

Testing note

The normal data above should be supplemented with a boundary case and an invalid or unusual case where the program includes validation. Expected output should be calculated before the program is run.

Pascal reference implementation

```
program TourBookings;
var
  i, bookingCount, adults, children: integer;
  customerName, packageCode: string;
  adultRate, childRate, amountPaid, cost, discount, finalCost, balance, totalValue: real;
begin
  totalValue := 0;
  write('Number of bookings: '); readln(bookingCount);
  for i := 1 to bookingCount do
    begin
      write('Customer name: '); readln(customerName);
      write('Package code: '); readln(packageCode);
      write('Adults: '); readln(adults);
      write('Children: '); readln(children);
      write('Amount paid: '); readln(amountPaid);
      if packageCode = 'CUL' then begin adultRate := 6000; childRate := 2500; end
      else if packageCode = 'ADV' then begin adultRate := 7000; childRate := 4000; end
      else begin adultRate := 7000; childRate := 3000; end;
      cost := (adults * adultRate) + (children * childRate);
      if adults + children >= 5 then discount := cost * 0.08 else discount := 0;
      finalCost := cost - discount;
      balance := finalCost - amountPaid;
      totalValue := totalValue + finalCost;
      writeln(customerName, ' balance: ', balance:0:2);
    end;
  writeln('Total booking value: ', totalValue:0:2);
end.
```

Documentation order

- Cover sheet
- Table of contents
- Problem definition
- Algorithm
- Source code
- Trace table and test data
- Program execution screenshots
- Final review

Final section check

- Every question in the practice brief has a visible completed result.
- The output uses the same project facts as the other sections.
- The work has been tested or checked rather than assumed correct.
- This SPARK example is watermarked and is for study only. Build your own teacher-assigned SBA.

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